



JobQuest

An AI-native experiential career-exploration platform
“Try the job before you pick the career.”

Chapters 1–5.1 · Corrections and remaining work · 26 September 2026

Specialized Project (ĐACN) · HCMUT

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Agenda

Five parts: corrections first, then the work not yet presented

A

Chapters 1–3

Corrected since
18 September

B

Chapter 4

Corrections, new
sections, remaining
work

C

Chapter 5.1

Analysis: process and use-
case models

D

Research

Research problems and
evaluation

E

Close

Summary and asks

CORRECTED

presented on 18 Sep, changed since

NEW

did not exist before

REMAINING

existed, not yet presented

Unchanged slides are not repeated

This deck reports Chapters 1 to 5.1.

What changed since 18 September

Every comment, where it is answered: the eight comments from the 18 September meeting

Feedback (18 Sep)	What we changed	Slide
References too old	Foundational works kept, each paired with recent evidence; the MBTI claim corrected	6, 7
Remove out-of-scope	Scope stated positively; the exclusion list removed	9
Objectives must give value to users	One overall objective plus four specific ones, each stating the user benefit	8
Scientific significance: stress AI and edtech	Reframed around AI in education	10
Business model and context	Public-benefit model, starting point and roadmap, organisation	25–27
Business rules without system constraints	Twelve business rules; six design constraints kept separate	21–24
Social cognitive aspect?	Sources of self-efficacy; screen versus live comparison	13, 20
Trim the research problems	Two AI research problems; the other three relocated	39–40

Your four questions of 23 September

Where each question is answered

Question (23 Sep)	What we changed	Slide
Who are the experts, and how are they linked?	Expert standard; experts work in the system on the occupations assigned to them	17, 36, 44
Why Cohen's κ , and what value is acceptable?	Quadratic weighted kappa with ETS acceptance thresholds	43
What does “task” include?	Nine formats and how each is scored	19
Is the evaluation feasible?	Two-stage evaluation, with sizing and workload	41–44



A

Chapters 1–3

Problem, evidence base and related work,
corrected since 18 September

The decision is made blind

People choose an occupation without knowing what the work is.

1 The obstacle is informational

In Gati, Krausz and Osipow's validated taxonomy, insufficient occupational information is a principal category of career decision difficulty, distinct from indecisiveness and from poor self-knowledge (**Gati, Krausz & Osipow, 1996**). Its structure was recently confirmed across 13 countries (**Levin et al., 2023**).

2 What guidance offers instead

Interest and personality inventories elicit what a person believes about themselves. They do not supply information about the work, so they cannot address that category of difficulty.

The evidence on MBTI, both sides

Weeks

Early retest studies: many respondents re-typed within weeks (**Pittenger, 2005**)

0.81–0.86

Publisher data: scale reliability (**Myers & Briggs Foundation, 2023**)

The real limit is what is measured: a stable preference is still a preference, not information about the work.

Two implications: the deficit that most needs fixing is informational, and the dominant instruments are structurally unable to fix it.

Changed since 18 Sep: MBTI evidence presented on both sides; Gati now paired with the 2023 cross-country confirmation.

Honest previews of a job improve fit

Established, not speculative: four decades of meta-analytic evidence.

1 Phillips, 1998: the effect

Meta-analysis of 40 studies. Realistic job previews reduce inflated expectations, increase self-selection, improve performance and lower voluntary turnover. Richer audiovisual previews outperform written ones **(Phillips, 1998)**.

2 Earnest et al., 2011: the mechanism

Meta-analytic path analysis over 52 studies and about 17,000 participants identifies perceived honesty as the mediating mechanism. The effect depends on candour, not on information volume **(Earnest, Allen & Landis, 2011)**.

No meta-analysis has since superseded it.

The most recent umbrella review identifies it as the current synthesis **(Graef, 2020)**.

The constraint was never evidence, it was delivery. Previews required workplace exposure, or content authored one occupation at a time.

Changed since 18 Sep: shown that the 2011 meta-analysis is still the current synthesis; its age reflects the field, not an outdated choice.

Objectives: value to the user

One objective stated as value to the user, four ways of reaching it

Overall objective

Enable a person facing an occupational decision to experience the work of an occupation, be assessed on how they performed it, and use that assessment to decide which occupations to explore next, before committing to one.

Four specific objectives: the means, and the benefit to the user

1 Faithful occupational tasks, with AI generation constrained by expert-authored material → **accurate information about the work**

2 Validated automated assessment → **a result, feedback, and guidance the user can rely on**

3 Competence converted into guidance on adjacent occupations → **exploration that builds on demonstrated ability**

4 Free, expert-governed access → **access that does not depend on ability to pay**

1-2 are the research problems; 3-4 are realised through design and the business model.

Changed since 18 Sep: objectives rewritten around user value: one overall objective, four specific ones, each stating the user benefit.

Scope, stated positively

Complete in function; coverage bounded to the IT field.

Why IT

IT is the only field where we can build an expert reference set, so our claims can be tested.

The design is domain-agnostic; the evaluated instance is IT.

What the work comprises

- Expert knowledge elicitation and review
- AI scenario generation
- Automated assessment
- Competence-based progression
- Engagement design
- Orientation conversation
- Catalogue and model administration
- Offline evaluation

Changed since 18 Sep: the out-of-scope list removed; scope stated positively, as what the work comprises.

Why this is worth doing

Two kinds of significance, justifying two different things.

Practical

Free, experiential occupational information before commitment, for students, graduates, and career changers. It rests on three established findings:

- Honest previews improve fit (**Phillips, 1998; Earnest et al., 2011**)
- Missing occupational information is a principal decision difficulty (**Gati et al., 1996; Levin et al., 2023**)
- Authentic task performance is the strongest source of self-efficacy (**Lent et al., 1994; Byars-Winston et al., 2017**)

Offered free, through public education authorities: reach does not depend on ability to pay.

Scientific: AI in education

- 1 **Grounded generation of educational content:** expert-in-the-loop, evaluated against a domain reference set
- 2 **Validated automated assessment:** chance-corrected, difficulty-stratified validation of LLM judging in a stealth-assessment setting
- 3 **A new edtech application:** experiential career education, where existing systems are small-sample research prototypes

Changed since 18 Sep: scientific significance positioned in AI-in-education and edtech research; free access added to practical significance.

Choices as trade-offs

Not right or wrong: situational judgment tests, the established method behind our choice tasks

1 What they are

Work situations with response options that differ in effectiveness: a “low-fidelity simulation”

(Motowidlo, Dunnette & Carter, 1990).

2 The evidence

102 validity coefficients, 10,640 people: validly related to job performance, $p = .34$ **(McDaniel et al., 2001).**

Items measure distinct constructs: interpersonal skill, teamwork, leadership **(Christian et al., 2010).**

3 How they are built

Experts supply critical incidents **(Flanagan, 1954)** → options are written from real behaviour → a different group of experts rates each option's effectiveness to form the key **(Whetzel et al., 2020).**

Design consequence: each option carries an expert-rated effectiveness profile across competencies, so a choice can trade one skill against another.

Career theory: fit is not a fixed verdict

Four traditions, and two of them say something most career systems ignore.

1 Holland

Occupational choice as person–environment congruence. Underlies the interest inventories in widespread use, including the public-domain instrument distributed with O*NET (**Holland, 1997**).

2 Social cognitive career theory

Self-efficacy, built principally through mastery experience, determines interest formation and choice: accessible to a system where users perform tasks, inaccessible to one that only elicits self-description (**Lent et al., 1994**). → **How self-efficacy forms, and what we can build: next slide.**

3 Super

Vocational self-concept develops over time, so fit is changing rather than fixed (**Super, 1980**).

4 Krumboltz

Unplanned events are a normal and potentially productive part of career development, not disruption (**Krumboltz, 2009**).

Consequence. Super and Krumboltz jointly justify two features: probabilistic life events (FR-14), and treating work–life balance as an attribute of an occupation rather than an externality (the five-attribute rating, FR-11).

Changed since 18 Sep: the social cognitive mechanism is now analysed in full (slide 13).

What we can build, and cannot

Social cognitive theory: answering “is there a social cognitive aspect?”

The four sources of self-efficacy (**Bandura, 1997**), and how JobQuest supplies each:



Mastery experience

Direct: the user performs the task



Vicarious experience

Little: no peers to observe



Persuasion / feedback

Through feedback and guidance



Affective state

Not targeted

Evidence

Mastery is the strongest source: meta-analysis (**Byars-Winston et al., 2017**); extended to career exploration (**Lent et al., 2017**); confirmed in a three-wave study of 1,512 students (**Chang et al., 2025**).

Limit

Self-efficacy is task-specific. Interaction is text-based, so we build confidence for written analysis, prioritisation, and decisions, not for oral fluency or interviewing. → **slide 20**.

Generation and judging

LLMs can judge open responses, but judging must be validated in four ways.

1 Feasibility

Agreement with human preferences above 80%, comparable to inter-human agreement, while the same work documents position, verbosity and self-preference bias (**Zheng et al., 2023**).

2 Method

Structured chain-of-thought against an explicit rubric improves alignment, with a residual preference for model-generated text (**Liu et al., 2023**).

3 Validation

Raw agreement overstates chance-corrected performance: a ~41-point gap between exact-match agreement and kappa on one benchmark (**“Reliability without validity”, 2026**). On difficult pairs with objective labels, the strongest evaluator reached ~64% (**Tan et al., 2025**).

The four requirements

✓ Report chance-corrected agreement: quadratic weighted kappa (slide 43) → **evaluation design**

✓ Control option ordering and response length → **assessment pipeline**

✓ Draw the evaluator from a different model family than the generator → **DC-01**

✓ Report performance stratified by item difficulty → **reference-set strata**

Changed since 18 Sep: the generator/assessor separation is now design constraint DC-01; the agreement metric is QWK.

B

Chapter 4

The proposed system: corrections, new sections, and the work not yet presented



What JobQuest is, and who it serves

Occupations as performable tasks; competence assessed from the performance.

Three presentation conventions, each chosen against evidence

RPG progression. Levels earned by demonstrated competence. Role-play shows the strongest genre-level association with skill acquisition: overall effect ≈ 0.82 , largest on skill acquisition (**“Effectiveness of the role-play method”, 2025**).

Visual-novel branching scenes. Narrative is among the strongest moderators in the gamification meta-analyses.

Roguelike career and life events. Narrative variety only: explicitly not coupled to any reward schedule or monetisation, per the harm evidence on randomised rewards (**“Randomized reward mechanisms”, 2023**).

Four user groups

Explorer. The end user; everything else serves them.

Domain expert. Senior practitioner on one or more occupations: co-authors content, play-tests, validates scenarios.

Content administrator (business administrator). One per occupation: co-authors with its experts; publishes and retires it.

System administrator (IT administrator). Creates occupations, assigns staff; models, coverage, cost.

Many-to-many: an occupation can have several experts; an expert can serve several occupations.

Changed since 18 Sep: responsibility per occupation made explicit: exactly one content administrator and one or more domain experts, who co-author its content.

Domain experts and how they link

System users, assigned per occupation, working alongside its content administrator

1 Who

- Senior or lead practitioners: about 5+ years in the role, experience supervising juniors
- 3–4 per role; 3 is the accepted minimum for content validation (**Lynn, 1986**)
- From at least two organisation types: product, outsourcing, startup, in-house
- About 22 IT roles → 66–88 experts, reached progressively

2 How: the link

Assigned to an occupation by the system administrator

Co-author context and rules with its content administrator

Play-test before review; practice runs excluded from metrics

Validate generated scenarios: approve, reject or correct

3 When

- Heaviest when seeds are first prepared, then annual revalidation
- Part-time, through partner organisations
- **Present phase: none.** The team prepares provisional seeds from public sources under advisor guidance; these are not claimed as expert-validated
- Stage 1: first panel of about 9 (3 pilot roles × 3)

Independence rule: a reviewer cannot approve content they initiated.

One mission, end to end

Experts and the content administrator author the context; the model expands it.

- 1 Role and context.** Backend Engineer, starter level. Your team, your stack, and tonight you are on call.
- 2 Scenario context (co-authored by domain experts and the content administrator).** The incident skeleton, the required facts, the keyed options, the rubric.
- 3 AI expands.** Specific symptoms, narrative and tone. Varied each run.
- 4 Choice point.** “Payments are down at 2 a.m. Where do you look first?” Branches the story. Each option is keyed to an expert-rated effectiveness profile: a trade-off, not a right answer.

- 5 Structured task.** Order the triage steps. Scored against the expert consensus ranking, with partial credit.
- 6 Free-response task.** Write the incident summary for your team. Scored against the rubric by a panel of judging models.
- 7 Result.** Score, response-specific feedback, competencies evidenced (debugging, communication), and credit toward the next level.

Hints available; hint use and response time are recorded as evidence.

Only the free-response step is a research risk. That is what research problem RP-2 is about.

Changed since 18 Sep: scoring now reflects trade-off keys and a judging panel; the context is co-authored by experts and the content administrator.

What a task includes

One scenario, one response, in one of nine formats

A task is one scenario within an occupation level, with a role, team and context, asking for one response. Its seed declares the competencies it targets.

Format	Scored by	In RP-2?
Multiple choice	Expert effectiveness profile per option	No
Multiple selection	Sum of selected profiles; harmful combinations flagged	No
Ordering	Distance from the expert consensus ranking; partial credit	No
Matching	Answer key if factual, otherwise an effectiveness key	No
True / false	Answer key; used sparingly	No
Timed choice	Option profile, plus on-device response time	No
Free text	Rubric, scored by the judging panel	Yes
Simplified email	Rubric	Yes
Simplified code	Tests (complete or debug) / rubric (explain or review)	Explanations and reviews only

Every option must be: grounded · plausible · intentional (a declared trade-off) · consistent with the scenario state · distinct. The model may rephrase an option but not change its keyed profile (DC-06).

What a screen task can represent

Functional task alignment matters more than realism (Hamstra et al., 2014)

Competency	Alignment with live work
Technical analysis	HIGH
Written communication	HIGH
Prioritisation and decisions	MODERATE
Adaptability	MODERATE
Collaboration	PARTIAL
Time pressure and stress	LOW
Oral communication and interviews	NOT ADDRESSED
Physical skill	NOT ADDRESSED

Alignment is highest for the core of IT work, which is itself screen-based: another reason for IT first. Results are never presented as readiness for unaligned competencies. Voice interaction is a candidate extension.

Twelve business rules (1/3)

And where each comes from: several are Chapter 2 findings made operational (report Table 4.1).

ID	Rule	Origin
BR-01 progression	The entry level of every occupation is accessible without prerequisite.	Exploration is the purpose; gating discovery defeats it
BR-02	Access to a higher level requires the competence credit specified for that level.	Competency-based progression (§2.2)
BR-03	Entry to an adjacent occupation requires demonstrated competencies overlapping that occupation's entry requirements, determined from the occupational taxonomy.	§2.2, §2.6
BR-04 integrity	An occupation may be rated only by a user who has completed it in full.	Rating validity depends on exposure

Changed since 18 Sep: the assessor rule was a technical constraint and is now DC-01; four rules are new, BR-09 to BR-12 (slide 23).

Twelve business rules (2/3)

And where each comes from (report Table 4.1).

ID	Rule	Origin
BR-05	No scenario context or grading criteria are published without domain-expert validation.	§2.5; accuracy of occupational representation
BR-06 ethics	Assessment results are formative. They are not transmitted to employers and do not determine access to any external opportunity.	§2.7; proportionality to assessment reliability
BR-07	Occupational representation must include the unfavourable aspects of the work.	Realistic job preview evidence (§2.1)
BR-08	Statistics derived from user performance are published only in aggregate form.	§2.7

Business rules are domain policies that hold whatever the technology. Implementation constraints are kept separately → slide 24.

Changed since 18 Sep: the assessor rule was a technical constraint and is now DC-01; four rules are new, BR-09 to BR-12 (slide 23).

Twelve business rules (3/3)

New since 18 Sep: responsibility per occupation, the replay cap, staff play-testing.

ID	Rule	Origin
BR-09 new	Each occupation is managed by exactly one content administrator.	Clear accountability for catalogue content
BR-10 new	Each occupation may be supported by one or more domain experts; the relationship is many-to-many.	Complex occupations may need more than one expert's judgment
BR-11 new	Within one scenario, the credit a user can gain toward a competency is capped; replaying cannot exceed the cap.	No unlimited progress from repeating one scenario
BR-12 new	Staff practice attempts made while testing a scenario before review are excluded from every explorer metric.	Keeps authoring-time testing out of real usage data

Six design constraints

The system-side limits, kept apart from business rules (report Table 4.8)

ID	Constraint	Rationale
DC-01 was a business rule	The assessor comes from a different model family than the generator	Self-preference bias
DC-02	Replaceable model provider	Portability
DC-03	Vietnamese personal data protection compliance, including minors	Legal obligation
DC-04	Enforced per-session cost ceiling	Non-revenue operation
DC-05	Browser-based, no installation	Universal access
DC-06 new	The generator may rephrase an option but not change its keyed profile	Validity of situational judgment scoring

A public-benefit platform, free for all

Community first, and the absence of revenue protects honest previews

1 Value

Free for every user; access gated by competence, never by payment.

The users who most need occupational information, students choosing under exam and family pressure, are the least able to pay.

2 Partners and funders

The Ministry of Education and Training (after city departments) and edtech organisations can each be both partner and funder. Industry partners and associations supply experts; HCMUT hosts.

Safeguards for all: no role in authoring, validating, or ranking content · aggregate data only · no charges, ads, or monetisation · all relationships disclosed.

3 Costs

Variable: model inference per session (NFR-11 monitored, DC-04 ceiling).

Fixed: expert authoring per occupation, which is why coverage grows field by field.

Levers: a cheaper panel of smaller judges ([Verga et al., 2024](#)), batch-generated and pre-reviewed scenario pools, and provider portability.

No commercial revenue means no incentive to portray jobs favourably: a safeguard, not just an access policy.

Where we start, and how we grow

Five stages, each unlocked by evidence

Policy context. The national career-guidance scheme (Decision 522/QĐ-TTg) ran 2018–2025 and was reviewed in November 2025, with a successor decree in preparation; Ho Chi Minh City targets $\geq 35\%$ of students in science, engineering, and technology by 2030.

Stage	When	Gate
0 University project	Now to end of 2026	Working system + Stage A targets met
1 Capstone and first pilot	2027	RP-1 and RP-2 thresholds met on pilot roles
2 Incubation, multi-site pilot	2027–28	Multi-site evidence; expert coverage of IT roles
3 Public-sector partnership	2028–29	City department first, then the Ministry → a formal agreement
4 National public resource	2030+	Sustained operation

Starting point: a university-hosted project that needs no partner and no funding beyond model costs, exactly where we are now. The research thresholds are the evidence we bring to funders.

Who runs it: a staged organisation

The three of us now; five units later (report Table 4.3)

Now (Stages 0–1): the team holds every role; the advisor provides academic oversight. **Later:** a steering committee (Ministry, university, funders, project lead) and an expert advisory board.

Unit	Positions	Responsibility
Product & Engineering	5	Platform, AI models, cost, security
Content & Occupational Expertise	Content lead, rubric specialist, part-time domain leads, external experts	Elicitation, review packages, revalidation
Research & Evaluation	2	Reference set, evaluation
Partnerships & Outreach	2	Ministry, schools, grants, expert recruitment
Data Protection & Compliance	1	Consent, minors' data

About 11 full-time positions, sized to grant funding. Content is governed per domain, each with a lead. IT domains: software engineering · data & AI · networks & infrastructure · information security · IT product & business analysis.

49 requirements, 6 constraints

Each requirement carries a business and a technical verification method.

Class	Total	Breakdown
FR	24	Explorer 14 · expert 5 · content admin. 2 · system admin. 4
NFR	12	Performance, reliability, privacy, usability: 2 each · 4 others
DR	13	7 from the system · 6 from sources, experts, explorers
Total	49	Paraphrased on slides 29–32

Where we think the risk concentrates

FR-04 generation keeps required facts · **FR-05** free-text scoring · **FR-07** capped competency recording · **FR-09** adjacency offers · **NFR-01/02** latency at the 95th percentile

One row, shown in full

FR-05 Score free-text responses on discrete anchor points (positive / neutral / negative), each supported by a quotation drawn from the response.

Business: domain experts judge a sample of scored responses fair and well-evidenced.

Technical: agreement with expert judgment on the reference set (quadratic weighted kappa, by difficulty); every quotation found verbatim in the response.

Ask. Priorities are not yet assigned in the updated specification. Which requirements do you consider essential for this phase?

Functional requirements (1/2)

FR-01 to FR-12, paraphrased from the updated specification; all are the explorer's

ID	Functional requirement	Group
FR-01	Sign up and sign in; minors need guardian consent	Explorer
FR-02	Open any published occupation's entry level, no prerequisite	Explorer
FR-03	Show the role, team and context before every scenario	Explorer
FR-04	Generate scenarios from the authored context, keeping required facts	Explorer
FR-05	Score free text on anchor points, each backed by a quote	Explorer
FR-06	Give result, feedback and guidance as separate parts	Explorer
FR-07	Record competencies shown, capped per skill in a scenario	Explorer
FR-08	Award credit on level completion and unlock levels	Explorer
FR-09	Offer adjacent occupations when competence overlaps enough	Explorer
FR-10	Optional orientation quiz ranking five candidate occupations	Explorer
FR-11	Five-attribute rating after full completion; show the aggregate	Explorer
FR-12	Private record of occupations, competencies and awards	Explorer

Functional requirements (2/2)

FR-13 to FR-24: explorer, domain expert, content and system administrators

ID	Functional requirement	Group
FR-13	AI-written user profile and social-orientation index	Explorer
FR-14	Random career and life events that shape later play	Explorer
FR-15	Co-author context and generation rules with the content admin.	Domain expert
FR-16	Approve, reject or correct generated scenarios	Domain expert
FR-17	Report where automated and expert judgments disagree	Domain expert
FR-18	Play-test before review, outside explorer metrics	Expert, content admin.
FR-19	Write reference grading criteria with AI help, validated	Domain expert
FR-20	Take one's occupation from draft to publication to retirement	Content admin.
FR-21	Create an occupation; assign one content admin. and ≥ 1 expert	System admin.
FR-22	Report catalogue coverage and last-validated dates	System admin.
FR-23	Configure generation and assessment models; they must differ	System admin.
FR-24	Monitor usage, cost and error rates	System admin.

Non-functional requirements

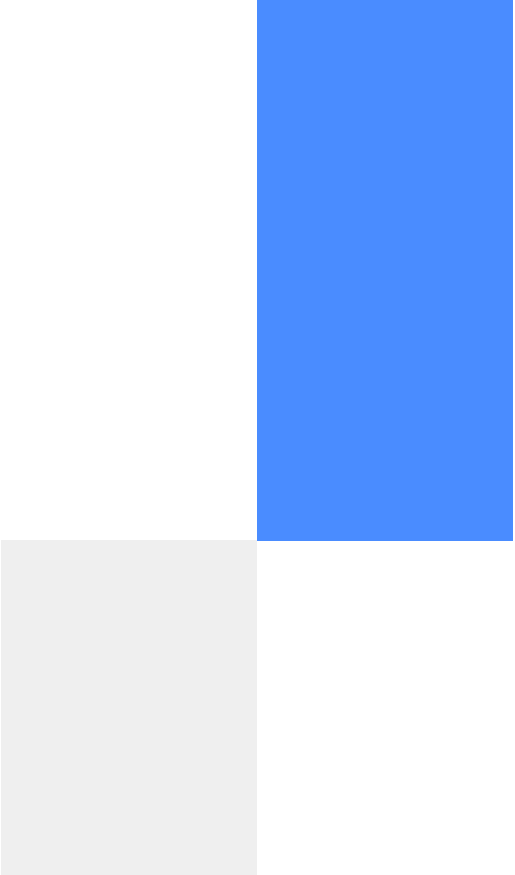
Each with a measurable target where one applies

ID	Non-functional requirement	Category
NFR-01	Generate a scenario in ≤ 5 s (95th pct.); show progress after 2 s	Performance
NFR-02	Assess a free-text response in ≤ 8 s (95th percentile)	Performance
NFR-03	Progress survives interruption and resumes without loss	Reliability
NFR-04	On provider failure: no progress lost, user told within 10 s	Reliability
NFR-05	Minimal personal data, stored encrypted, deletable on request	Privacy
NFR-06	Withhold aggregate statistics below a minimum population	Privacy
NFR-07	Tell users that scenarios and assessments are machine-made	Transparency
NFR-08	Interface and content in Vietnamese and English	Usability
NFR-09	$\geq 80\%$ of readers can state a task's objective after reading it	Usability
NFR-10	Add a new occupation without changing code	Maintainability
NFR-11	Mean model cost per session under the ceiling; cost logged	Cost
NFR-12	Swap the model provider without editing authored content	Portability

Data requirements

Where each data set comes from

ID	Data requirement	Source
DR-01	Occupations, tasks, work activities and skills	O*NET, ESCO (adapted)
DR-02	Competency taxonomy, soft and technical, with indicators	From DR-01
DR-03	Adjacency between occupations, with overlap scores	Computed from DR-01/02
DR-04	Scenario context: facts, competencies, skeleton, rules	Content admin. + experts
DR-05	Generated scenarios, traced to context and model version	System
DR-06	User profile and progress: credit, levels, history	System
DR-07	Assessment records: score, competencies, feedback, guidance	System
DR-08	Five-attribute ratings, only after full completion	Explorers
DR-09	Expert reference set, gold standard, stratified by difficulty	Domain experts
DR-10	Telemetry: latency, cost, errors, model versions	System
DR-11	Per-scenario credit cap for each user and competency	System
DR-12	Flag separating staff practice runs from real sessions	System
DR-13	AI-written user profile and social-orientation index	System



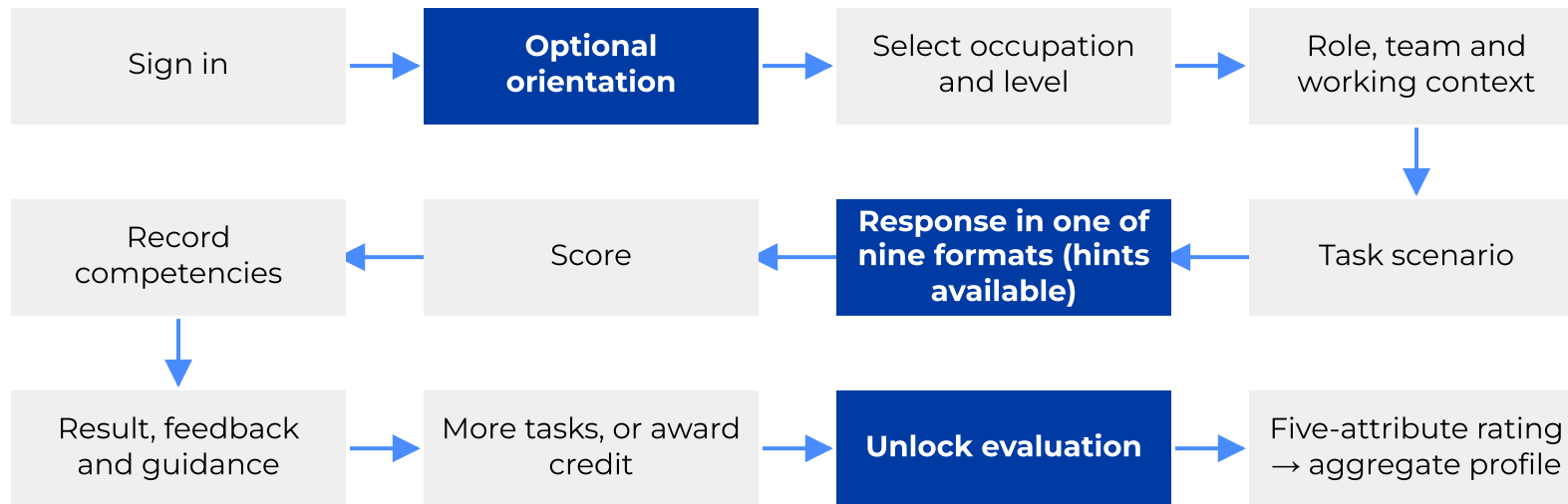
C

Chapter 5.1

Analysis: the process models and the use cases

The exploration process

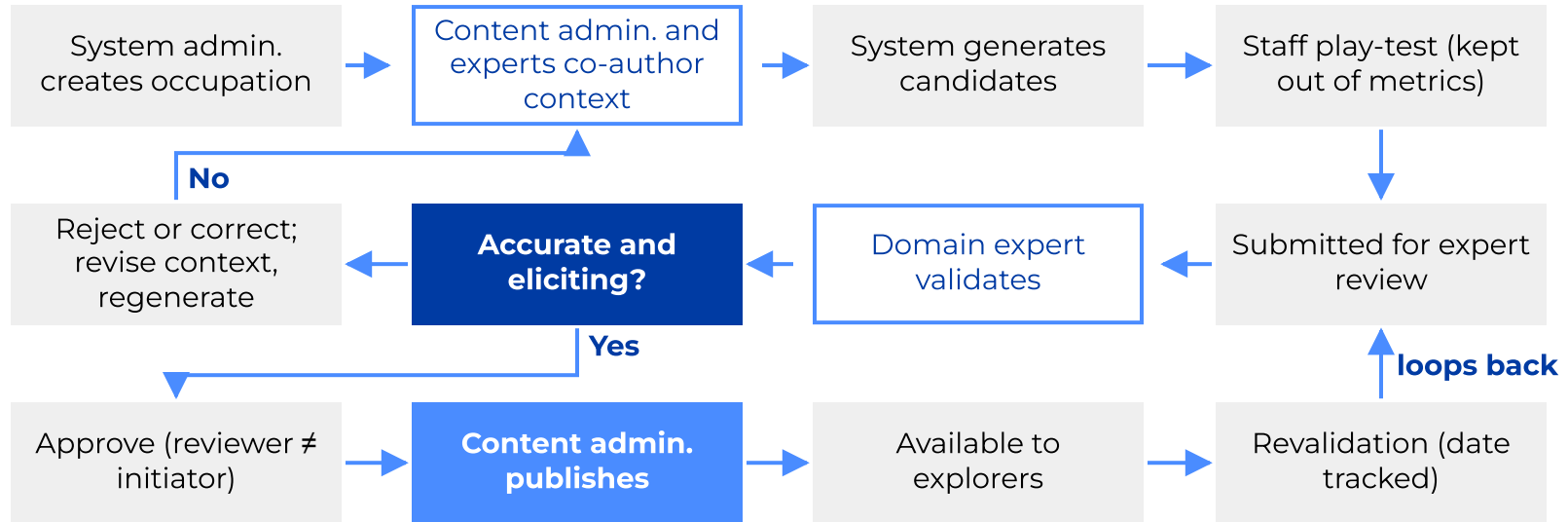
The primary user-facing process, with three decision points.



Navy boxes are the three decision points: the **optional orientation** (deliberately not a prerequisite, since making it one would turn us back into an assessment instrument), the **response format** (expert effectiveness key, answer key, or rubric), and the **unlock evaluation** (where BR-02 and BR-03 apply). Full diagram with every branch: report Figure 5.1.

The content lifecycle

Content administrator and domain experts co-author; experts validate: BR-05 made operational.



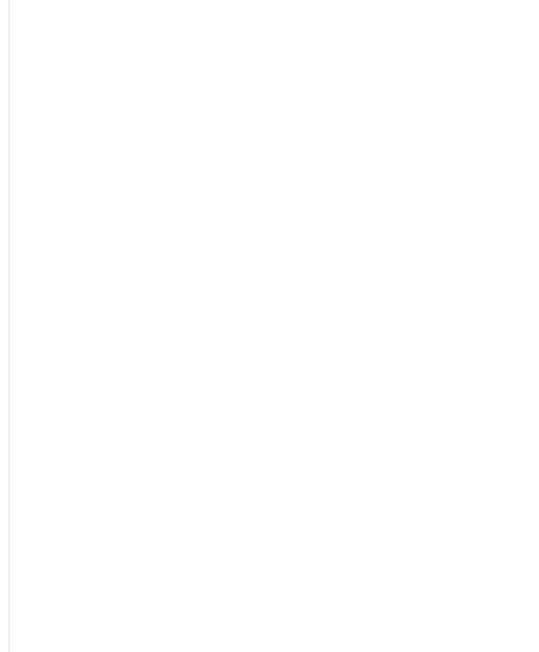
The rejection path branches. Experts may reject one scenario or correct the context; a corrected context regenerates every prior generation from it. Each decision is recorded with the reviewing expert.

Where the expert constraint lives. Experts co-author the context, play-test, validate scenarios, review assessment disagreements and author reference items. Their time is why phase 1 is one field.

Use cases (1/2)

Explorer and domain expert; the two use cases in bold carry the research.

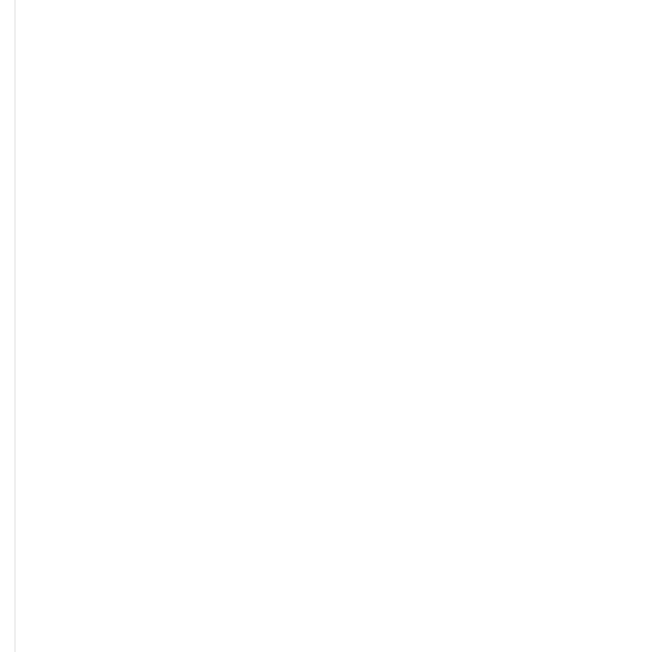
 Explorer use-case diagram



Explorer

- UC-01
Explore the catalogue
- UC-02
Orientation quiz
- **UC-03
Perform a task
(RP-2)**
- UC-04
Progress further
- UC-05
Rate an occupation
- UC-06
Personal

 Domain expert use-case diagram



Domain expert

- UC-07
Author context and rules*
- **UC-08
Validate a scenario*
(RP-1)**
- UC-09
Review disagreement

Use cases (2/2)

Content administrator (one per occupation) and system administrator.

 Content administrator use-case diagram

 System administrator use-case diagram

- UC-07 Author context and rules*
- **UC-08 Validate a scenario***
- UC-10 Occupation lifecycle

System administrator

- UC-11 Coverage and currency
- UC-12 Configure models
- UC-13 Operation and cost
- UC-16 Create occupation

UC-08 exception: a scenario that can be passed without its target competency sends the context back for revision, and a reviewer cannot approve content they initiated.

14
Test-

D Research and evaluation

Two research problems, and how we will
answer them



Two research problems

Both in AI applied to education; each passes three tests of what a research problem is

Three conditions: the answer is not already known · it is knowledge that generalises beyond this system
· our evaluation produces evidence that answers it.

RP-1: grounded generation

“When generation is constrained by an expert-authored seed specifying required facts and target competencies, what proportion of generated scenarios do domain experts judge to be occupationally accurate and to require the target competency, and how does this compare with generation from the same instructions without the seed?”

The comparison makes the answer knowledge.

RP-2: validity of LLM stealth assessment

“To what extent does rubric-based automated judgment of task responses agree with domain-expert judgment of the same responses, measured by quadratic weighted kappa, relative to the agreement between experts themselves, and how does that agreement vary with item difficulty and across repeated runs?”

Expert-expert agreement is the ceiling.

Three former RPs, now in their place

Kept, but in their right place

Former RP	What it asked	Why it is not a research problem	Now in
RP-3	How to map demonstrated competence to occupational adjacency	A design decision based on taxonomies; no new knowledge	Chapter 5 design
RP-4	Which game mechanics sustain engagement without displacing learning	Answerable only by longitudinal user study	Future work
RP-5	How to build the expert reference set	The method for answering RP-1 and RP-2, not a problem	Chapter 7 evaluation

How we will prove it: two stages

Two stages, because experts arrive in stages

A Stage A: now (DACN)

- Synthetic, criterion-perturbed benchmark
- Panel of judging models from different families
- Tests sensitivity, robustness, consistency
- **Cannot** show agreement with experts on real responses, and we do not claim it

B Stage B: capstone (complete product)

- Real responses from consenting volunteer participants
- Generated scenarios
- Rated independently and blind by 3 experts per pilot role
- Answers RP-1 and RP-2 in full

Why generation and assessment are reported separately

Three distinct failure modes: biased judgment, unstable judgment, and judgment failing selectively on hard items. Each is diagnosable only when reported separately; a single accuracy figure hides which one occurred.

Stage A: the synthetic benchmark

Ground truth by construction, before any expert is available

1 Perturbation

A strong reference response, plus variants that each degrade exactly one rubric criterion to a known level, negative to positive: ground truth by construction (**Sai et al., 2021**).

2 Judging panel

Several judges from different model families score independently and repeatedly. A panel of smaller diverse models outperformed a single large judge, with less self-bias and at lower cost (**Verga et al., 2024**). Synthetic responses come from a family none of the judges belongs to.

3 Guarding against circularity

- Bias probes (length padding, reformatting, reordering) should not change scores
- The advisor checks the designed levels on a random ~20% sample

Illustrative size: 3 roles × 5 free-text tasks × 4 criteria × 4 levels ≈ 240 responses, plus bias probes (estimate).

What counts as success

Established thresholds, not ones we chose

Why quadratic weighted kappa

Chance correction is why we use kappa at all: raw agreement overstates performance. Rubric scores are ordinal: plain Cohen's κ treats a 1-point and a 3-point disagreement alike; QWK weights larger disagreements more.

RP-1 criterion	Threshold	RP-2 criterion	Threshold
Fact preservation	$\geq 95\%$ of seed-required facts (proposed target)	QWK, automated vs human	≥ 0.70
Content validity	I-CVI $\geq .78$ per scenario; S-CVI/Ave $\geq .90$ per role (Polit, Beck & Owen, 2007)	Standardised mean difference	≤ 0.15
Seeded vs unseeded	Seeded > unseeded, McNemar's test $p < .05$	Degradation from human-human to automated-human agreement	≤ 0.10 : exactly what RP-2 already asks
Expert agreement	Krippendorff's $\alpha \geq .800$ (Krippendorff, 2004)	RP-2 thresholds: the ETS framework (Williamson, Xi & Breyer, 2012).	

Reported per role and per difficulty stratum; a failure on hard items is reported, not averaged away.

Is the evaluation feasible?

Answering your question with numbers

Stage B, per expert (estimate)	Items	Time
RP-1 scenario ratings	48 × ~4 min	~3.2 h
RP-2 response ratings	50 × ~4 min	~3.3 h
Option keying	40 × ~1 min	~0.7 h
Total: about one working day		≈ 7 h

Stage B at a glance

- Panel: 3 pilot roles × 3 experts = 9
- ~144 scenarios (3 roles × 8 seeds × 2 conditions × 3 samples)
- ~150 real responses (3 roles × 50) from consenting HCMUT IT-student volunteers

Where responses come from: volunteers on the completed product, or participants reached through partner organisations. Until then, Stage A needs no human responses.

Summary, next steps, and the asks

Chapters 1 through 5.1 are complete and internally consistent.

Delivered

- Evidence base (Ch 2), now including situational judgment tests and measurement standards
- Related work with derived criteria (Ch 3)
- System specification: 12 business rules, 6 design constraints, 49 requirements, business model and roadmap (Ch 4)
- Process models and 16 use cases across 4 actors (Ch 5.1)

Next

- Finalise the ~22 IT roles and choose the 3 pilot roles
- Build the Stage A benchmark
- Recruit the Stage 1 expert panel through partners
- Implement seed entry and the review-package workflow
- Complete Ch 5.2–5.3 (technology and design)
- Re-verify the Vietnamese market claim close to submission

The asks

- 1 Business context: starting point, roadmap, organisation (25–27)
- 2 RP-1 and RP-2 with QWK and the acceptance criteria (39, 43)
- 3 The expert model and evaluation feasibility (17, 44)
- 4 Criteria set and ratings, if not yet confirmed (presented 18 Sep)
- 5 Requirement priorities for this phase (28–32)

“Try the job before you pick the career.”